

Focal Plane Array using VCSELs for Beam Steering in High-Speed Indoor Optical Wireless Communication

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Abstract We demonstrate an indoor optical wireless link using a focal plane array of 2×2 VCSELs as the multi-user transmitter. Each VCSEL can individually be addressed by driving FPGA. Four channels of maximum 10 Gbps OOK each can be used to realize wireless link error-free. ©2025 The Author(s)

Introduction

Data generation and consumption is growing rapidly due to several factors, such as the proliferation of the Internet-of-Things, the increase in popularity of streaming services, the adoption of artificial intelligence cloud computing, and the digitalization in all services [1]. However, the increase in requested wireless data has strained the radio frequency (RF) technologies, are limited by the available bandwidths and regulated frequency spectra [2]. To address this problem, Optical Wireless Communication (OWC) technologies have been proposed that offer numerous advantages over traditional RF-based wireless technologies. These include huge bandwidth, lower latency, enhanced privacy and security, and minimal interference to and from neighboring wireless networks because light cannot go through walls [3]. Our approach does not seek to replace the traditional RF networks, but to complement them. By offloading user services needing high bandwidth and low latency services to OWC, we can realize a congestion-free and high-performance network for indoor users.

For optimal OWC performance, the system must deliver maximum possible power to receivers while maintaining eye safety compliance. This necessitates narrow, precisely steered beams that efficiently direct power to receiver apertures. Vertical-cavity surface emitting lasers (VCSELs) for OWC offer several advantages because of their vertically surface emitted light and compact size and therefore can easily be arranged in a 2-dimensional structure for beam steering. Current VCSELs have demonstrated large bandwidth for direct intensity modulation that makes them suitable for indoor OWC use cases [4].

While various techniques already exist for steering narrow optical beams: polarization gratings [5], arrayed waveguide grating routers [3], metasurfaces [6], optical phased arrays [7], and spatial light modulators [8]. Our proposed concept we use a 2-dimensional VCSEL array for generating modulated light beams and for steering them to discrete directions equal to the

number of VCSELs. In this paper, we employ a 2×2 VCSEL array where individual emitters can be selectively activated to create beams in 4 specific directions. A key advantage of this design is that multiple VCSELs can operate concurrently to simultaneously serve different users with independent data streams.

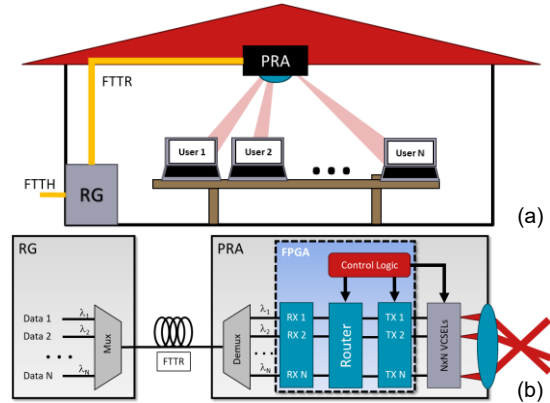


Fig. 1: Indoor OWC connecting fiber-to-the-home (FTTH) via residential gateway (RG) to pencil beam radiating antenna (PRA) using fiber-to-the-room (FTTR): (a) Schematic and (b) Components making up RG and PRA.

Fig. 1a illustrates the schematic of an indoor OWC infrastructure where FTTR forms a broadband optical fronthaul of PRAs. Multiple PRAs can be installed on the room ceilings to maximize the coverage area and to overcome non line-of-sight conditions. Fig. 1b shows optical and electrical components for RG and PRA. The RG has user specific wavelength, with the channels multiplexed onto a single fiber and demultiplexed on the ceiling. into separate inputs of the FPGA. This FPGA electrically switches the data streams to the appropriate VCSELs in the array, which is positioned at the lens's focal plane for precise beam steering.

Our VCSEL array features a $250\text{-}\mu\text{m}$ pitch between adjacent emitters and $350\text{ }\mu\text{m}$ between rows, creating a compact light source at the 1300nm band. The beam steering angles are determined by the activated VCSEL relative position to the focal axis. This can be modelled through paraxial ray analyses. Considering an aberration-free thin ideal lens represented by the matrix

equation 1 [9]:

$$\begin{bmatrix} x_1 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/f & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ \theta_0 \end{bmatrix} \quad 1$$

where x_0 and x_1 represent the initial and final ray positions, θ_0 and θ_1 denote the initial and final ray angles, and f is the lens focal length. This paraxial approximation maintains accuracy for angles of up to approximately 10° , sufficient intended indoor application scenarios. To optimize performance across indoor scenarios, we selected a couple of spheric lenses with focal lengths of 4, 6, and 9 mm, enabling us to customize the angular coverage for specific scenarios. The focal lengths result in the maximum steering angles of 3.1° , 2° , and 1.4° relative the optical center. The total field of view is determined by utilizing a pair of VCSELs working in complementary fashion.

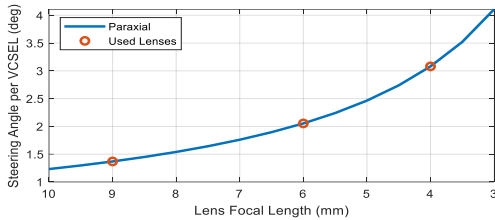


Fig. 2: Comparison between the paraxial equation and lenses

Fig. 2 demonstrates how the focal length affects the beam steering performance. For applications requiring even larger wider angles, it requires further reducing the focal length of the steering lens optimized for the operating wavelengths or increasing the size of the VCSEL array.

Experiments

We focus exclusively on the PRA performance, isolating the wireless transmission characteristics while ensuring optimal data feed to evaluate receiver signal quality. This setup (see Fig. 3) uses commercial available VCSEL driver controlling the activation and modulation of individual VCSELs in the 2×2 array. Each VCSEL operates in the O-band (1300nm) producing a maximum optical power of 0 dBm with a 3 mA bias current and 7.8 mA modulation current, more clearly shown in Figure 3.

The beam-steering element consists of a spherical N-BK7 lensed (diameters of 2mm for EFL of 4mm and 3 mm for EFL of 6mm and 9mm) with selectable focal lengths as previously described. The lens aligns to the 2D array center, positioning each VCSEL $215 \mu\text{m}$ from the optical axis.. Data transmission is accomplished through dedicated transceivers on the FPGA for each VCSEL. The system employs 100Ω differential signalling with impedance-matched traces. Signal integrity is maintained via 100Ω differential probes connecting to the VCSEL driver.

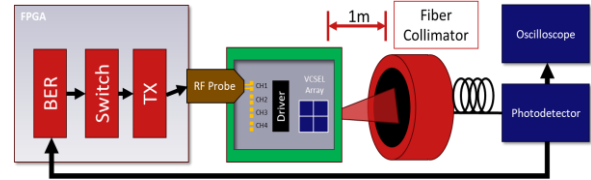


Fig. 3: Block diagram of experimental setup

The receiver used is an adjustable fiber collimator coupling the light into a $50\text{-}\mu\text{m}$ multi-mode fiber, connected to a 25-GHz TIA amplified photoreceiver. Since the main point is to characterize the transmitter of a wireless link, we are employing a wide bandwidth receiver to avoid any receiver-related performance limitation. Despite having a narrow field-of-view, this receiver setup allows us to have a higher bandwidth than the existing wide field-of-view wireless receivers.

The system control implements a broadcast-and-select approach through an FPGA that efficiently manages data switching while simultaneously generating pseudo-random binary sequence patterns for transmission. Upon reception, the FPGA analyses the returned signals to calculate bit error rates. Communication with the VCSEL driver occurs via I2C protocol, enabling adjustment of the VCSEL currents and output.

Results and Discussion

The performance of our focal plane array (FPA) beam steering was characterized by using three measurements: eye diagrams, bit-error-rate, and steering angle.

For eye diagrams, we used non-return-to-zero on-off keying (OOK) signal formats. We observed the eye diagrams of electrical driving signals and the corresponding VCSEL outputs. All 4 VCSELs were evaluated using a high-bandwidth sampling scope, with the receiver at 1m from the array, as depicted in Fig. 3. Figure 4 shows the eye diagrams at 10 Gbps for each channel in the 2×2 VCSEL array. Channel 1 displays an eye height of 650 mV and eye width of 100 ps, with excellent opening despite its lower output power, 4 dB less compared to the other channels at 0 dBm. Channel 2, due to its optimal position close to the driver, as shown on the VCSEL layout on Fig. 4, demonstrates superior signal quality with the cleanest eye opening and minimal jitter. Channel 3 exhibits performance characteristics comparable to Channel 1, with adequate eye opening for reliable data transmission. Due to the position of Channel 4, it exhibits increased timing jitter and noise because of the mismatch between the differential signal paths from the driver. Despite this, Channel 4 shows a clear eye opening also. When comparing the FPGA RF signal with the corresponding optical outputs, we observe that the driver circuit substantially enhances signal quality

through its equalization and amplification stages before the signal being used to modulate the VCSELs.

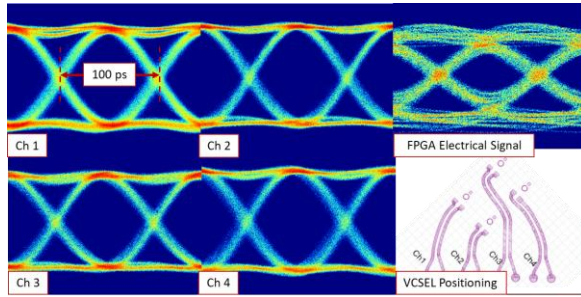


Fig. 4: Eye diagrams from the four VCSEL channels operating at 10 Gbps. FPA is driven by FPGA electrical signal (top-right).

Following the eye pattern, we evaluated the transmission reliability through link budget across all four channels. This measurement was conducted by adjusting the received optical power by defocusing the fiber collimator and measuring the corresponding BER at each power level, with the same setup used for the eye-diagram with the receiver at 1m from the array.

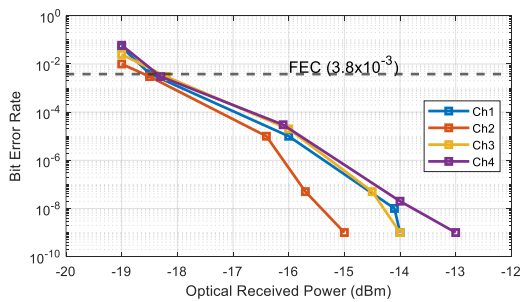


Fig. 5: BER performance vs optical power for the four VCSEL channels at 10 Gbps. The horizontal dotted line denotes the FEC threshold. BER was taken 1m from the PRA.

Figure 5 depicts the BER versus received optical power relationship for all four VCSEL channels at 10 Gbps. All channels show that with increasing optical power, the BER reduces to error-free, with Channel 2 demonstrating superior sensitivity (-18.5 dBm at FEC threshold) due to its array position closer to the driver. Channels 1 and 3 present comparable performance curves with approximately 0.5 dB lower receiver sensitivity at 10^{-9} than Channel 2, while maintaining consistent performance across their spatial positions. Channel 4 displays the most pronounced performance degradation, which aligns with the amplitude jitter and reduced eye opening observed in its eye-diagram in Figure 4. This degradation is attributed to the impedance mismatch.

Fig. 6 presents a comparative analysis of the steering angles achieved using three focal length lenses (4 mm, 6 mm, and 9 mm). These measurements were conducted at a transmission distance of 70 cm from the VCSEL array.

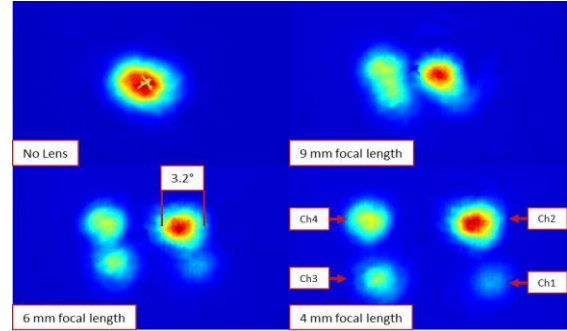


Fig. 6: Beam steering performance for different focal length lenses, displaying achieved angular displacement vs theory.

The experimental results demonstrate that the 4 mm focal length lens provides the widest steering angle of 3° , while the 6 mm and 9 mm lenses deliver progressively narrower steering angles of 2° and 1.4° respectively, agreeing with the expected values. Each beam maintains a beam waist around 3.2° , which translate to 50mm diameter beam at 1m distance, across all configurations, ensuring a narrow beam to the receiver. Importantly, despite the relatively close spacing of the VCSEL elements, the inherently narrow beam characteristics combined with the smaller focal lengths produce well-defined, spatially separated beams with no visible crosstalk between channels. This spatial isolation is particularly advantageous for multi-user scenarios where interference between data streams must be minimized.

Conclusions

This paper presents a VCSEL-based 2D beam steerer for an OWC transmitter in the 1300nm wavelength range. The FPA-based PRA has an ability to generate narrow beams, with 3.2° divergence angle, that can be steered towards desired users. We demonstrate 4×10 Gbps transmission for a 1m wireless distance. The BER performance for all channels can be made error-free with a less than 2dB performance variation across the VCSELs at $\text{BER}=10^{-9}$. Our experimental results confirm that varying focal lengths of steering lenses significantly impact the quality of the beams and the achievable steering angles. The compact $250 \times 350 \mu\text{m}^2$ source provides a high-performance OWC implementation using commercially available parts. When positioned at the focal plane of steering lenses with different focal lengths from 4mm to 9mm, the PRA can have maximum steering angles from 3.1° to 1.4° , offering flexibility for coverage requirements. This work shows that VCSEL-based FPAs have the potential to be deployed in real indoor OWC systems to serve multiple users with 10Gbps/user and more. Due its small size, this FPA can also be applied to the user devices that require small space and low power (battery-driven) usage.

Acknowledgements

The authors thank Alight Technologies for the 1300-nm VCSEL array sample provided.

This work is financially supported by the Netherlands Organization for Scientific Research (NWO) in the national Perspective program under the grant number 12128 “Optical Wireless Super-highways: FREE Photons”. Also, the Growth funds PhotonDelta and 6G-Future Network Services are acknowledged for partial funding of the experiments.

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